

X	0	1
f	20	100

X	f	expected frequency f(x)=Np(x)	
0	20		5.642978112
1	100		46.1822771
2	190		165.3561596
3	280		338.319759
4	350		432.627743
5	300		354.0636506
6	220		181.1041229
7	120		52.93431324
8	3		6.768996445
Total	1583		

		Formula
N	1583	=B31
n	8	
np	4.04548326	=SUMPRODUCT(B22:B30,A22:A30)/B31
p	0.505685407	=B36/B35
q	0.494314593	=1-B37
P(X<3)	0.137196093	=BINOMDIST(2,B35,B37,1)
P(X<=3)	0.350916724	=BINOMDIST(3,B35,B37,1)
P(X>3)	0.649083276	=1-B40
P(X>4)	0.375787166	=1-BINOMDIST(4,B35,B37,1)
P(4<X<7)	0.611367983	=BINOMDIST(6,B35,B37,1)-BINOMDIST(3,B35,B37,1)

2	3	4
190	280	350

Formula	Rounded expected frequency	Formula
=\$B\$34*BINOMDIST(A22,\$B\$35,\$B\$37,0)	6	=ROUND(C22,0)
	46	
	165	
	338	
	433	
	354	
	181	
	53	
	7	
	1583	=SUM(E22:E30)

5	6	7	8
300	220	120	3